

User study of an AR reading aid system to promote deep reading

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ABSTRACT

We proposed a reading aid system combining AR HMD and occlusion technology to cultivate students' intensive reading habits and promote deep reading. A user study was conducted to explore the differences in reading comprehension and subjective evaluation between AR and VR mode. Analysis results show that there are statistically significant differences for realism factors in subjective evaluation between the two modes. The participants confirmed the usefulness of the AR system and preferred the setting of actual paper reading. These results indicate there is potential for AR reading aid system to be used as an educational aid to promote deep reading.

Keywords: Augmented reality, comparative study, deep reading.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/augmented reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies.

1 INTRODUCTION

Deep reading [1] is a concept that is opposite to shallow reading and emphasizes readers' gradual exploration of the deep connotation of the text. Generally the content of fragmented reading using electronic devices to read in piecemeal time is short and easy, which means people just browse the content and turn to next text quickly through hyperlinks. Influenced by such free and jumping reading style, high school students tend to be irritable and read out of order, which prevents students from deep reading. Despite the rapid development of digital reading, paper reading which is a kind of deep reading and a process of knowledge internalization is still essential for high school students. Therefore, it is significant for high school students influenced by fragmented reading to cultivate students' intensive reading habits and improve the ability of deep reading.

Compared with traditional learning methods, students' reading comprehension ability and learning persistence show a higher level in Augmented Reality (AR) environment. To the best of our knowledge, no research has been conducted to study the cultivation of students' reading habits in AR environment. We propose a reading aid system which combines AR HMD and occlusion technology to ensure that users read in order and conduct a user study with 20 high school students to explore the differences in reading comprehension and subjective experience between AR and Virtual Reality (VR) mode.

2 HARDWARE AND IMPLEMENTATION METHOD

The reading aid system proposed in the paper took Microsoft

HoloLens 2 as the main display device and used Unity3D to establish multiple equidistant virtual planes to occlude the text line by line. By using Vuforia AR SDK to identify the QR code on the upper left corner of the reading text, users can observe the virtual plane displayed above the paper and occlude the content of text. As the main purpose of the proposed reading aid system is to promote deep reading, students were requested to focus on reading instead of gesture interaction in HoloLens. Therefore, a Bluetooth keyboard was chosen as interaction interface. Users only need to click on the space bar to remove the virtual plane line by line. HoloLens was also the hardware in VR mode and its external view was blocked by an A4 paper to provide virtual immersive environment. The parameters of virtual planes and interaction method in VR mode were the same as AR mode.

3 USER STUDY

The first goal of the user study is to find out whether there are differences between AR and VR mode in reading comprehension. The second one is to find out which mode users prefer. The reading texts are literary readings selected from national college entrance examination in 2017 and 2018. The participants are 20 juniors (14 boys, 6 girls) from the High School Affiliated to Renmin University of China. Before the experiment, we conducted a short demographic survey including questions on prior experience with AR and VR systems and general reading habits. Students were given a brief introduction on how to start the reading program in HoloLens.

In order to check whether the order of the two modes has a significantly impact on reading comprehension and exclude memory problems caused by using the same reading texts, the experiment was designed as a crossover study [2] and students were randomly divided into 4 groups. Students of group 1 first read text 1 in AR mode and then read text 2 in VR mode. Group 2 was opposite to group 1 but the text in each mode remained unchanged. Students in group 3 first read text 2 in AR mode and then read text 1 in VR mode, whereas group 4 was opposite to group 3. After finishing reading in each mode, students answered 6 questions about the content of the reading text within 5 minutes. Scenes of students reading are shown in Figure 1.



Figure 1: Scenes of students reading in AR mode (left) and VR mode (right).

After completing reading comprehension tests, students were prompted to fill out questionnaire 1 which questions were grouped in the six factors proposed by Witmer to measure presence in virtual environments [3]: sensory factors (SF), control factors (CF), distraction factors (DF), ergonomic factors (EF), realism factors (RF), and other factors (OF). The questions listed in questionnaire

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1 are 7-scale Likert questions with 1 meaning *strongly disagree*, 2 *disagree*, 3 *somewhat disagree*, 4 *neutral*, 5 *somewhat agree*, 6 *agree*, and 7 *strongly agree*. After finishing reading comprehension tests in the second mode, students filled out questionnaire 2 whose questions were the same listed in questionnaire 1 with six additional subjective questions. The questionnaires and reading texts are in supplemental materials.

4 RESULTS

IBM SPSS 25 software was used to analyze experimental data consisting of the rate of reading comprehension and the subjective evaluation results of 6 factors in the two modes. For all of the analyses detailed hereafter, significance tests were two-tailed and conducted at the 0.05 significance level.

4.1 The rate of reading comprehension

The Shapiro-Wilk test shows p -value = 0.026 for text 1 and p -value = 0.011 for text 2, which means the data do not follow a normal distribution. Therefore, non-parametric Mann-Whitney U test was used for statistical hypothesis testing. For text 1, there are no statistically significantly differences between AR mode (mean rank = 10.2) and VR mode (mean rank = 10.80), $U = 47$, $Z = -0.237$, p -value = 0.853. For text 2, there are no statistically significantly differences between AR mode (mean rank = 11.45) and VR mode (mean rank = 9.55), $U = 40.5$, $Z = -0.756$, p -value = 0.481. There are two reasons which may lead to the results. The two modes assist students to read in order with unlimited reading time so that students fully understand the content in each mode. In addition, the high quality of literature of participants means the rate of reading comprehension only depends on the difficulty of the text rather than the systematic form of reading.

4.2 Questionnaire 1

The Shapiro-Wilk test shows that the data of SF (p -value = 0.181), CF (p -value = 0.067), DF (p -value = 0.342), RF (p -value = 0.280) and OF (p -value = 0.229) follow a normal distribution except for EF (p -value = 0.026). Therefore, Wilcoxon signed-rank test was used for the data of EF and the remaining five factors were tested using t-test for paired data. For the data of EF, Wilcoxon signed-rank test indicated $Z = -834$, p -value = 0.404. Other 5 factors results are shown in Figure 2 and only the data of RF shows statistically significant differences at 0.05 significant level. RF represents occlusion effect and practicality of the reading aid system. The average scores obtained in VR mode is higher than that in AR mode, which means students have a better reading experience in VR mode.

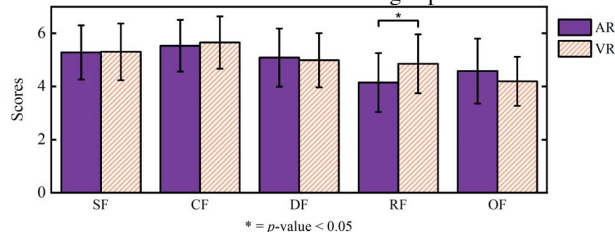


Figure 2: Scores of five factors in AR and VR mode.

4.3 Questionnaire 2

Questionnaire 2 raised three main questions about usefulness, preference and recommendation between AR and VR mode and the answers were shown in Figure 3. Most students first confirmed that reading in AR mode is useful for cultivating the habit of reading in order. Then they mentioned the disadvantages of the AR mode. On the one hand, some students commented that the virtual planes in AR mode didn't completely occlude the text especially in the edge

area. In other words, students observed the real text through the virtual planes. On the other hand, the field of view (FoV) of HoloLens is limited so that the virtual plane is incomplete from the student's perspective. Both of aspects are caused by inherent hardware and display properties of HoloLens. The FoV is only 30° [3] and images displayed on the screen are translucent. We set the material of the occlusion plane to be self-luminous and adjust the brightness as high as possible to obtain a better occlusion effect. The problem that followed is that brightness negatively affects students for long-term reading. There were also some students preferring to AR mode despite the disadvantages of the mode. They preferred to paper reading because it was natural and comfortable and hoped AR reading aid system can be further improved.

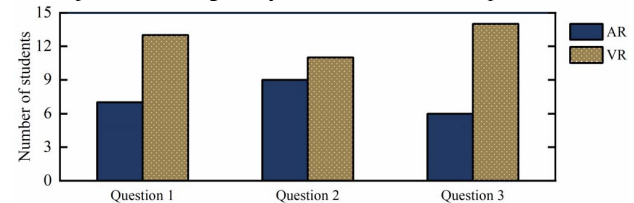


Figure 3: The number of students who chose AR and VR modes among the three questions.

5 CONCLUSION

Despite the rapid development of digital reading, paper reading is still essential for high school students who are influenced by fragmented reading. In this paper, we present an AR reading aid system to promote deep reading and conduct a user study with 20 high school students to explore the differences in reading comprehension and subjective experience between AR and VR mode. The analysis results show that there is no statistically significant difference in reading comprehension between the two modes, but there is a statistically significant difference in RF that measures occlusion effect and practicality of the two reading aid systems. Due to the limited FoV and inherent display properties of HoloLens, the occlusion effect of AR mode still needs to be improved in the future. The subjective answers in Questionnaire 2 also mentioned the issue of occlusion effect, but most students stated that the AR reading aid system is useful for reading in order and promoting deep reading. In addition, students like the setting of actual paper reading which make them natural and comfortable in the AR system.

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